

☒ Easy (10 Questions)

Q1. What is the purpose of a firewall?

- A) Increase network speed
- B) Filter network traffic
- C) Store data
- D) Create backups

Answer: B

☒ *Firewalls inspect and filter traffic based on security rules.*

Q2. What does DMZ stand for in networking?

- A) Dynamic Machine Zone
- B) De-Militarized Zone
- C) Digital Media Zone
- D) Data Management Zone

Answer: B

☒ *DMZ is a security buffer zone for public-facing services.*

Q3. A countermeasure in cybersecurity refers to:

- A) Data storage
- B) Mitigation of risk
- C) Virus spreading
- D) Backup creation only

Answer: B

☒ *Countermeasures reduce the likelihood or impact of threats.*

Q4. Which device filters traffic based on rules?

- A) Server
- B) Printer
- C) Firewall
- D) Modem

Answer: C

Q5. The area between two firewalls that hosts public services is called:

- A) Internal network
- B) VLAN
- C) DMZ
- D) VPN

Answer: C

Q6. Which of the following is NOT a firewall method?

- A) Three-legged
- B) Dual-firewall
- C) Physical-only
- D) None

Answer: C

Q7. Which risk treatment involves insurance?

- A) Accept
- B) Avoid
- C) Transfer
- D) Exploit

Answer: C

Q8. Which of the following is a **preventive** control?

- A) Antivirus
- B) Audit log
- C) Security camera footage
- D) Forensic toolkit

Answer: A

Q9. Risk = ?

- A) Threat + Exposure
- B) Vulnerability × Impact × Threat
- C) Damage / Time
- D) Likelihood × Uptime

Answer: B

Q10. What is an exposure in InfoSec?

- A) Antivirus software
- B) Degree of potential loss
- C) Hardware firewall
- D) Backup strategy

Answer: B

☒ Medium (20 Questions)

(Coming in the next message...)

Shall I continue with the **20 Medium** and **10 Hard** level MCQs?

You said:

yes

ChatGPT said:

You're giving feedback on a new version of ChatGPT.

Which response do you prefer? Responses may take a moment to load.

ChatGPT

Response 1

Continuing with the remaining:

☒ Risk Management, Exposure, Firewall, and DMZ – MCQs (continued)

☒ Medium (20 Questions)

Q11. Which of the following is a correct definition of risk in information security?

- A) The number of users in a system
- B) The probability of a threat exploiting a vulnerability
- C) The level of antivirus protection
- D) A type of firewall rule

Answer: B

☒ *Risk = Likelihood × Impact of a threat exploiting a vulnerability.*

Q12. In which layer of the OSI model do most traditional firewalls operate?

- A) Physical
- B) Data Link
- C) Network
- D) Application

Answer: C

☒ *Traditional firewalls inspect packets at the network layer (Layer 3).*

Q13. A company uses two firewalls with a DMZ between them. This design is known as:

- A) Split firewall
- B) Multi-path security
- C) Dual-firewall DMZ
- D) Triple-hop security

Answer: C

☒ *Dual-firewall DMZ architecture places public services between internal and external firewalls.*

Q14. Which type of control is a firewall categorized as?

- A) Administrative
- B) Technical and preventive
- C) Physical
- D) Recovery

Answer: B

☒ *Firewalls are technical controls, and their primary purpose is prevention.*

Q15. Exposure is best described as:

- A) A malware sample
- B) A way to track user login
- C) Potential for loss or damage
- D) The amount of physical RAM

Answer: C

Q16. Which of the following would NOT typically be placed in a DMZ?

- A) Web server
- B) Mail server
- C) Internal HR server
- D) DNS server

Answer: C

☒ *DMZ is only for public-facing services; HR server is internal.*

Q17. Packet filtering firewalls work by:

- A) Analyzing application-layer protocols
- B) Inspecting connection states
- C) Filtering based on IP address and port
- D) Using deep learning

Answer: C

Q18. A firewall that examines ongoing connections and their state is called:

- A) Stateless
- B) Stateful
- C) Hybrid
- D) Application

Answer: B

Q19. One of the main disadvantages of a single firewall DMZ is:

- A) Reduced latency
- B) Complexity
- C) Single point of failure
- D) Too many interfaces

Answer: C

☒ *If it fails, both LAN and DMZ can be compromised.*

Q20. Which firewall method offers better segmentation and security but is more expensive?

- A) Stateful inspection
- B) Dual-firewall implementation
- C) Proxy-based filtering
- D) NAT-based routing

Answer: B

Q21. Risk mitigation refers to:

- A) Accepting all risks
- B) Sharing risk with another party
- C) Implementing controls to reduce impact
- D) Ignoring minor threats

Answer: C

Q22. Which of the following best defines a countermeasure?

- A) A threat actor
- B) A method of deleting logs
- C) A protective action taken against a threat
- D) A type of attack

Answer: C

Q23. What is the purpose of the third interface in a 3-legged firewall?

- A) Monitor VPNs
- B) Host backup storage
- C) Connect to DMZ
- D) Encrypt internal traffic

Answer: C

Q24. A firewall that acts as a gateway between clients and services is a:

- A) Packet filter
- B) Proxy firewall
- C) Stateless filter
- D) Static NAT

Answer: B

Q25. Which of the following is most often used to assess exposure and impact?

- A) Subnetting
- B) VLAN tagging
- C) Risk assessment
- D) OS fingerprinting

Answer: C

Q26. In a risk formula, if a system has **no vulnerability**, then:

- A) The risk is high
- B) The risk is negligible
- C) Threats increase
- D) Exposure is irrelevant

Answer: B

☒ *No vulnerability = almost no chance for threat to succeed.*

Q27. Firewall rules are usually configured using:

- A) USB commands
- B) Graphical firewall DNS systems
- C) Access Control Lists (ACLs)
- D) DHCP configuration

Answer: C

Q28. A mail server placed in the DMZ is compromised. What's protected in a dual-firewall setup?

- A) Internal web servers
- B) DMZ services
- C) Internal network resources
- D) Antivirus databases

Answer: C

Q29. Which of the following is a disadvantage of firewall use?

- A) Hides logs
- B) Slows down login
- C) May block legitimate traffic
- D) Deletes untrusted files

Answer: C

Q30. Which is NOT a benefit of using a DMZ?

- A) Public access control
- B) Internal data isolation
- C) Automatic antivirus update
- D) Protection against external attacks

Answer: C

☒ Hard (10 Questions)

Q31. What is the main reason to use a proxy firewall?

- A) It works at the physical layer
- B) It enables deep packet inspection
- C) It increases download speed
- D) It reduces logging needs

Answer: B

☒ *Proxy firewalls can inspect traffic at the application layer in depth.*

Q32. Which situation best illustrates the need for risk transfer?

- A) Keeping old software
- B) Buying cyber insurance
- C) Installing antivirus
- D) Accepting policy violations

Answer: B

Q33. In quantitative risk analysis, **Single Loss Expectancy (SLE)** is calculated as:

- A) $SLE = ARO \times ALE$
- B) $SLE = ARO \times \text{Cost}$
- C) $SLE = \text{Asset Value} \times \text{Exposure Factor}$
- D) $SLE = ALE \div ARO$

Answer: C

☒ *SLE = expected monetary loss from a single event.*

Q34. What feature distinguishes NGFWs from traditional firewalls?

- A) IP filtering only
- B) Stateless inspection
- C) Deep Packet Inspection and threat intelligence
- D) Only outbound traffic control

Answer: C

Q35. What happens when a risk is accepted?

- A) It's eliminated
- B) It's ignored
- C) No controls are applied, but it's acknowledged
- D) It's patched

Answer: C

Q36. A DMZ is used in which type of network architecture?

- A) Flat network
- B) Zero Trust
- C) Segmented or tiered network
- D) Cloud-only environment

Answer: C

Q37. Which type of attack can a firewall **not** typically prevent?

- A) IP spoofing
- B) Denial-of-Service
- C) Internal threats
- D) Port scanning

Answer: C

☒ *Firewalls are perimeter defenses; they don't protect against insider threats.*

Q38. If an attacker gains access to a system due to a known vulnerability not being patched, the risk was:

- A) Avoided
- B) Transferred
- C) Mitigated
- D) Accepted

Answer: D

Q39. A firewall uses **stateful inspection**. What does it track?

- A) Total bandwidth
- B) MAC addresses
- C) The state of active connections
- D) File permissions

Answer: C

Q40. The most secure but costly firewall implementation is:

- A) Three-legged
- B) Dual firewall DMZ
- C) Software-based firewall
- D) Host-based only

Answer: B